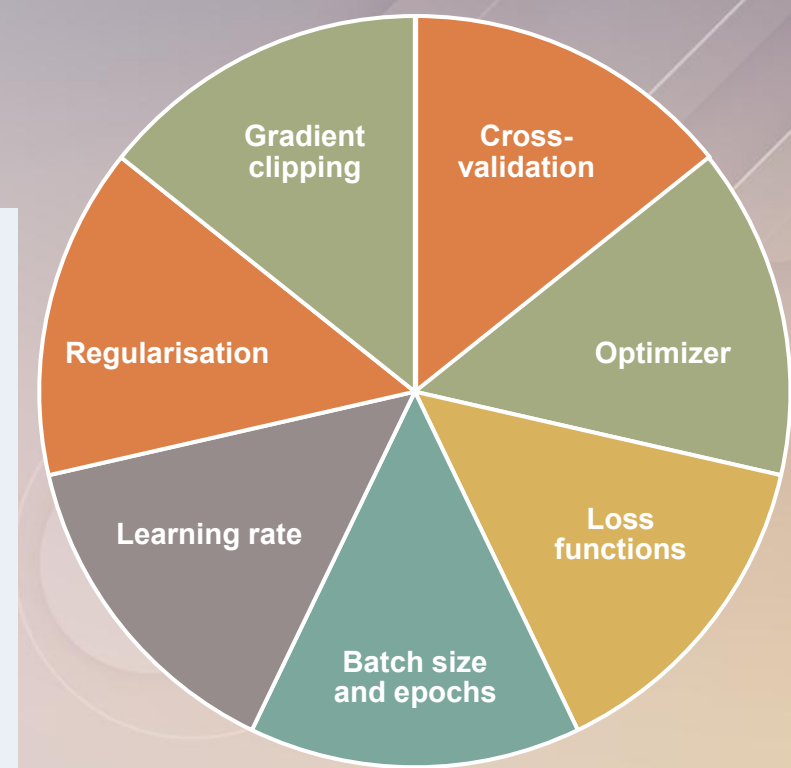


Team-discussion prompts: Sequential Models and Hyperparameter Tuning

Task for all teams:

1. Describe situations where you will use vanilla RNN instead of LSTM. How about GRU instead of LSTM (or vice-versa)? Why?
2. Take a problem that is suitable for either LSTM or GRU (any NLP application, e.g., classification, translation, summarisation, etc.)
3. Use any common dataset: suggestions include news classification dataset <https://keras.io/api/datasets/reuters/> or IMDB reviews https://keras.io/examples/nlp/bidirectional_lstm_imdb/
4. Choose **3 of the 7 hyperparameters** to explore (You can also try to use Pytorch instead of the sample notebook that uses Keras/Tensorflow)
5. Vary the chosen indicators to either high/mid/low (e.g., learning rate, batch size), or using different options (e.g., types of optimizers, loss functions) and explain the significance of the impact on downstream tasks. **Show the results and generate plots on your code.**
6. What happens if a “less appropriate” hyperparameter value or type is being used?



Other functions to explore within Python:

1. Early stopping Callback function: patience, verbose, mode, etc.
2. Model checkpoint
3. Filters/kernel size for GCN or any convolutional networks
4. Useful resource: **Optuna** (Bayesian optimization)
 - <https://optuna.org/>
 - <https://github.com/optuna/optuna>

General references:

- <https://towardsdatascience.com/simple-guide-to-hyperparameter-tuning-in-neural-networks-3fe03dad8594>
- <https://towardsdatascience.com/experiments-on-hyperparameter-tuning-in-deep-learning-rules-to-follow-efe6a5bb60af>
- <https://www.akshaymakes.com/blogs/experiment-tracking>
- <https://machinelearningmastery.com/how-to-grid-search-hyperparameters-for-pytorch-models/>
- <https://medium.com/analytics-vidhya/rnn-vs-gru-vs-lstm-863b0b7b1573>
- <https://ieeexplore.ieee.org/abstract/document/9599044>
- <https://towardsdatascience.com/illustrated-guide-to-lstms-and-gru-s-a-step-by-step-explanation-44e9eb85bf21>